

The Physics of Juggling

Gravity, Rotations, and General Relativity

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Contact Juggling

Newton's Laws of motion

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Newton's Laws of motion

1. Without force, objects keep moving.

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 - ▶ Inertia is quantified by **mass**.

$$F = ma,$$

Force = mass $\times$ acceleration.

Contact Juggling

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3. All forces exist in pairs.

Ball juggling

Motion under gravity

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$$F = -mg.$$

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Motion under gravity

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- ▶ Heavier objects experience a proportionally larger force.
- ▶ This force acts to increase the **downwards velocity** (i.e. accelerate downwards).
- ▶ This force does not affect the sideways velocity.

Ball juggling

Parabolic motion

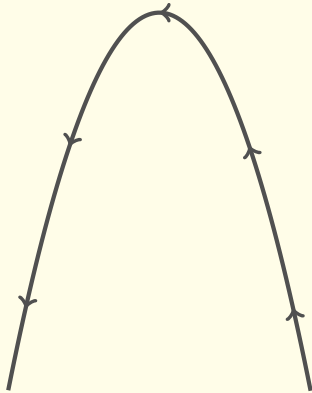
- ▶ The ball travels with constant speed sideways. . .
- ▶ . . . and with an increasing speed downwards
- ▶ This path is called a parabola

Parabolic motion

What's actually going on

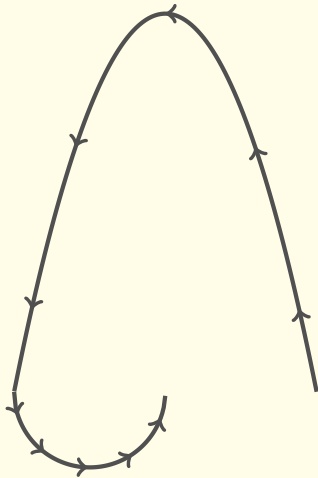
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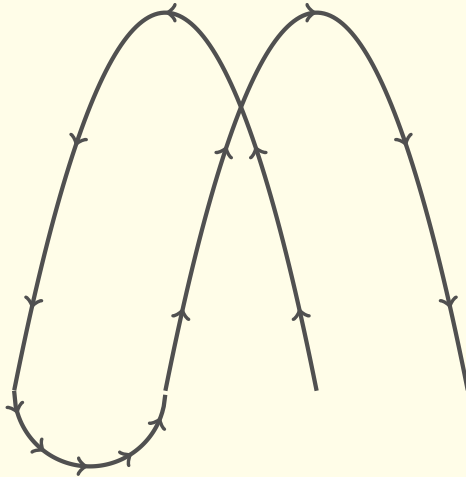
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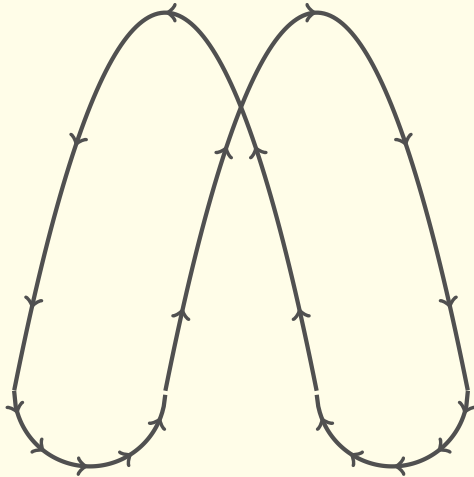
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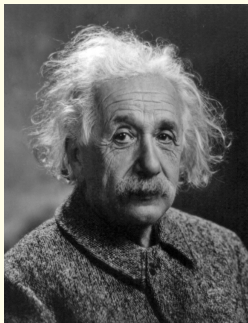
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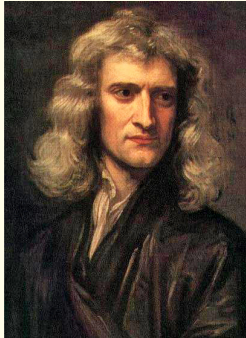
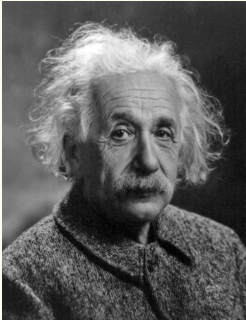
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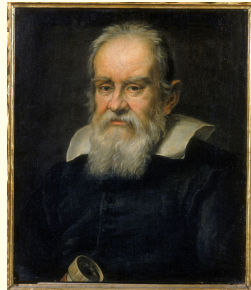
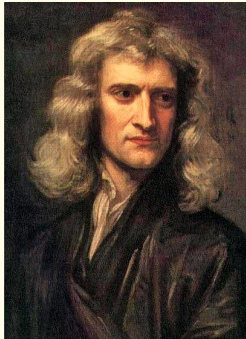
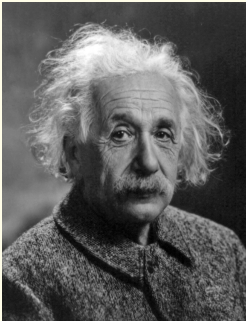
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Club Juggling

Angular momentum

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- ▶ Spinning objects have **angular momentum**.

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Torque $\neq$ Force

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Torque $\neq$ Force

- ▶ An object in free fall experiences no torque, so it remains spinning about an axis.

Angular momentum

Stability of rotations

Momentum depends on

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1. Speed

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Angular momentum depends on

1. The speed at which it is spinning.
2. The size, shape and mass of the object.
3. The axis about which it spins.

Diabolo

Gyroscopes

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Diabolo

Gyroscopes

- ▶ When an object is spinning very fast, it has a very large angular momentum.
- ▶ We call this a gyroscope: (like a spinning top)

Poi Spinning

Circular motion



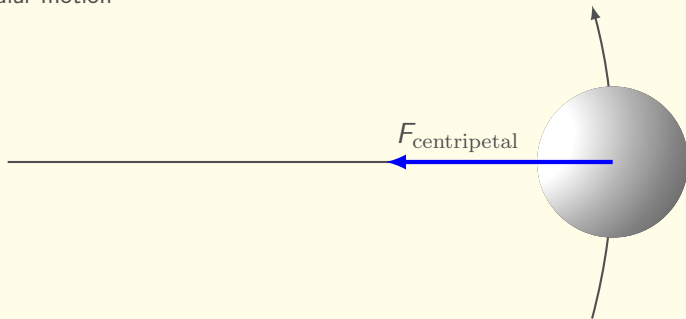


Poi Spinning

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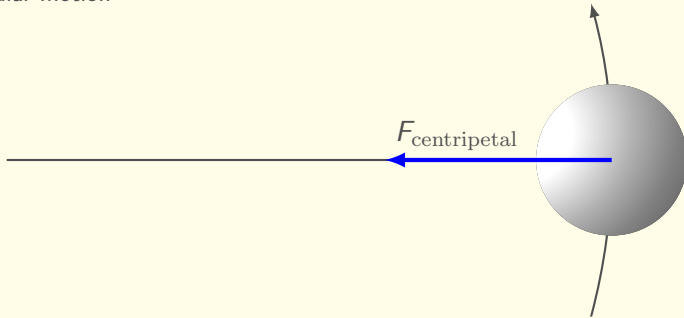
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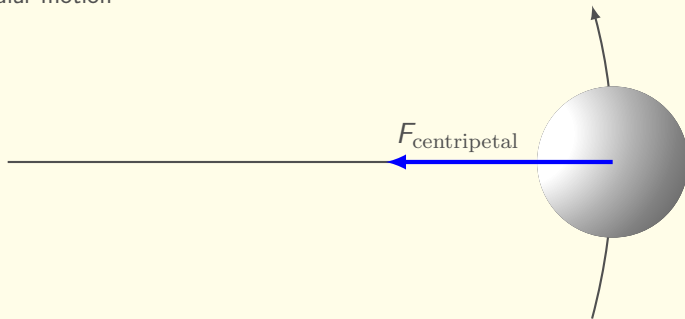
Circular motion



- Poi spinning is an example of **circular motion**.

Poi Spinning

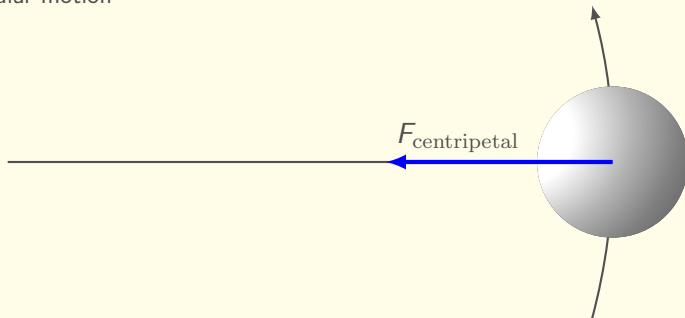
Circular motion



- ▶ Poi spinning is an example of **circular motion**.
- ▶ The weight has a continuously **changing** momentum.

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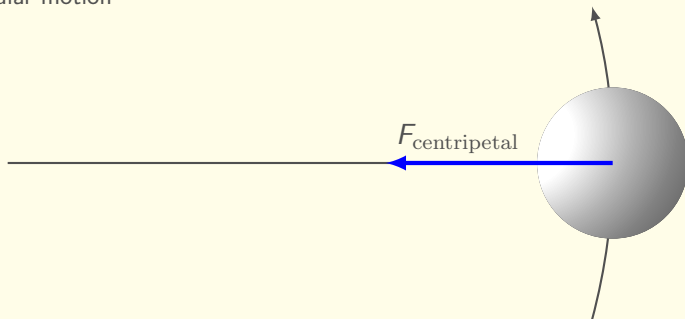
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Circular motion



- ▶ Poi spinning is an example of **circular motion**.
- ▶ The weight has a continuously **changing** momentum.
- ▶ Although the speed of the weight may remain constant, its momentum changes with the direction of travel.
- ▶ The force which provides this is **centripetal force**:

$$F_{\text{centripetal}} = -m \frac{v^2}{r} = -m \omega^2 r.$$

Poi Spinning

Centrifugal force vs. centripetal force



Poi Spinning

Centrifugal force vs. centripetal force



- If we were to rotate “with the poi”, it would appear stationary.

Poi Spinning

Centrifugal force vs. centripetal force



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- ▶ Two forces would appear to be acting on the weight.

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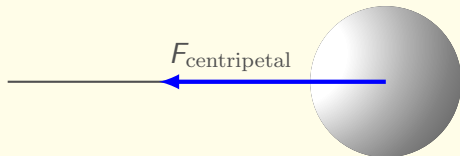
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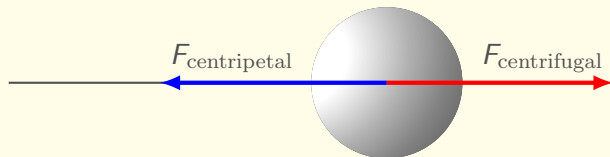
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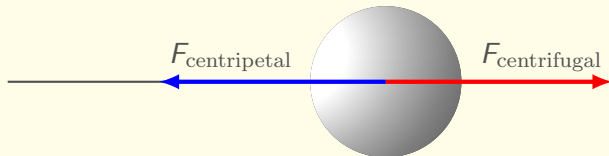
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- ▶ Two forces would appear to be acting on the weight.
 1. Centripetal force inwards: $-mv^2/r$
 2. Centrifugal force outwards: $+mv^2/r$
- ▶ A new force has arisen by changing our point of view

Fictitious forces

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- ▶ A classic sign that a force is fictitious is that it is **proportional to mass (m)**
- ▶ “Normal” forces (e.g. friction, electric & magnetic force) cannot be removed by a change of reference.

Cigar boxes

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Cigar boxes

Free fall

- ▶ Cigar box juggling operates in *free fall*.
- ▶ Because all objects fall at the same rate, **there is no gravity in free fall**.
- ▶ This is commonly observed (and used) by astronauts.

Free fall

Simulating zero-gravity



General Relativity

Heavier objects don't fall faster

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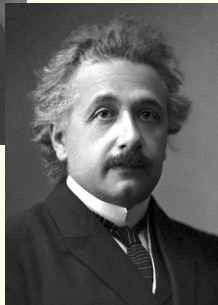
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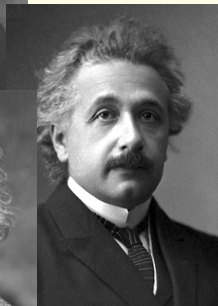
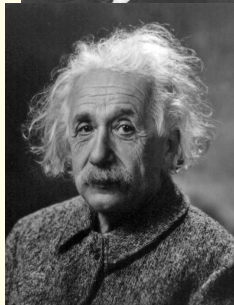
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Gravity as a fictitious force

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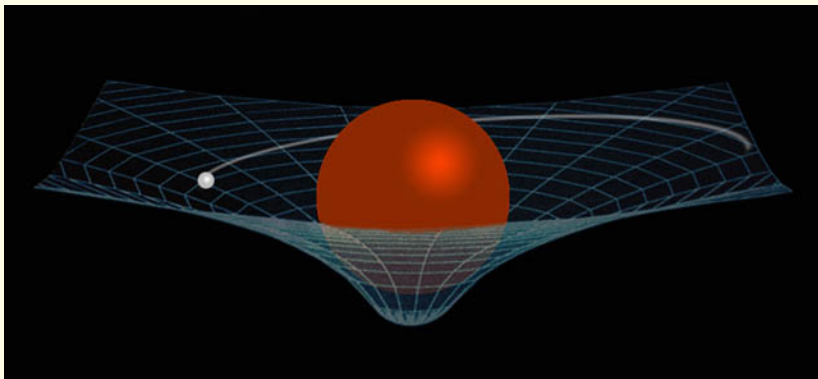
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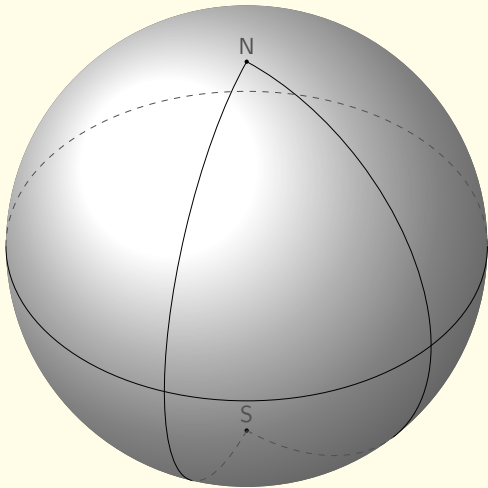
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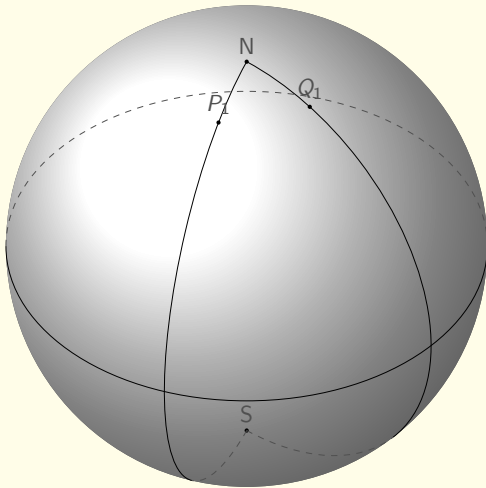
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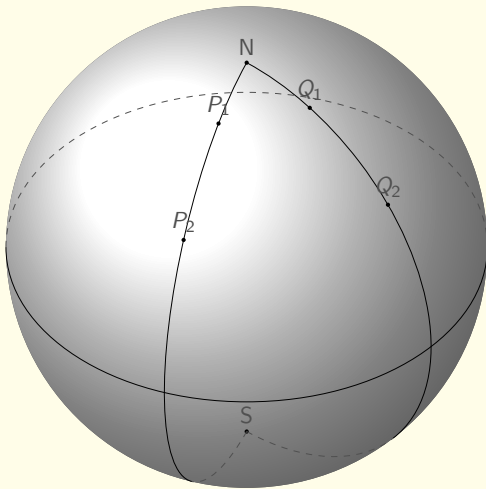
- Two planes flying away from the north pole



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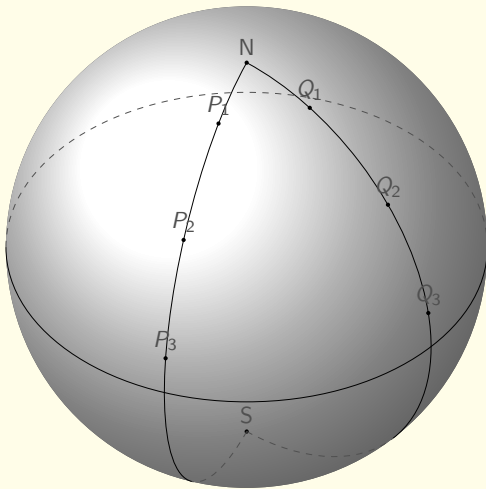
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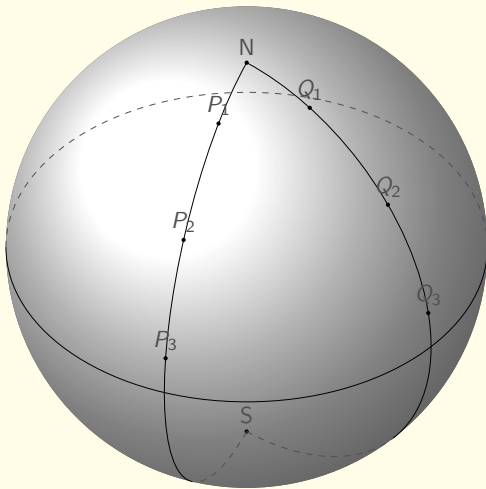
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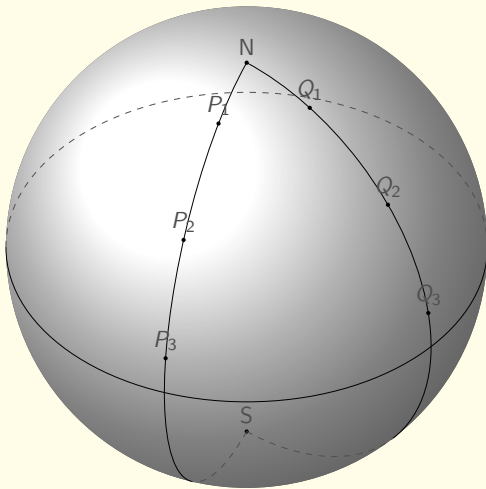
- ▶ Two planes flying away from the north pole
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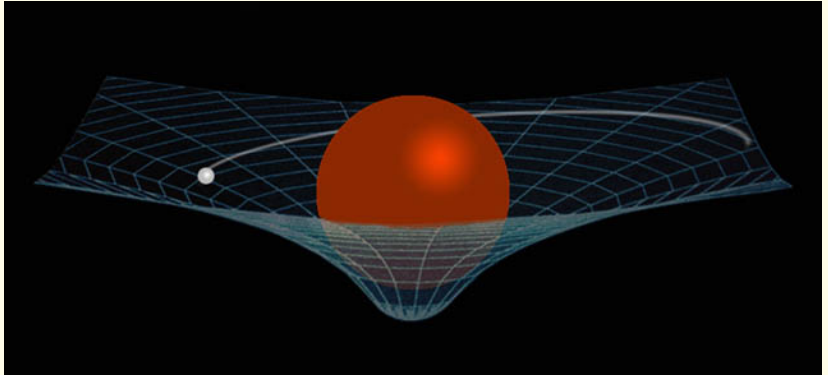
Gravity as a fictitious force

- ▶ Two planes flying away from the north pole
- ▶ They fly “in a straight line”
- ▶ But seem to be drawn together



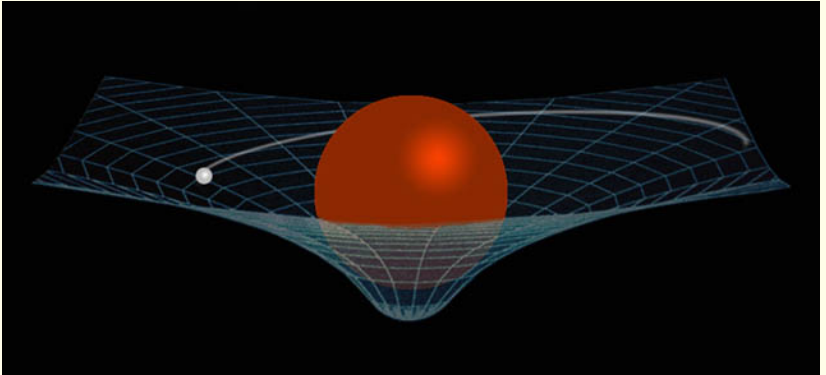
Space warps

The essence of general relativity



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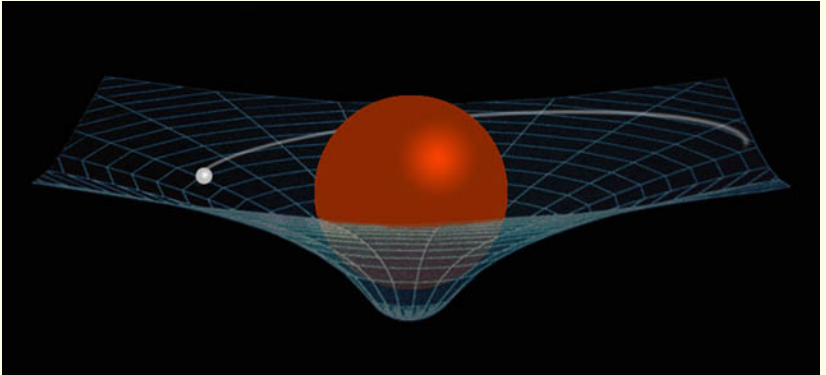
The essence of general relativity



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Space warps

The essence of general relativity



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- ▶ Matter tells space how to curve

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